Worksheet 3.1 — Derivatives of Sinusoidal Functions

Grade 12 Calculus and Vectors (MCV4U) · Unit 3: Derivatives of Transcendental Functions

The Big Ideas

In radians, $\frac{d}{dx}\sin x = \cos x$ and $\frac{d}{dx}\cos x = -\sin x$; combine with the chain, product and quotient rules.

- $\frac{d}{dx}\sin x = \cos x$, $\frac{d}{dx}\cos x = -\sin x$.
- Chain: $\frac{d}{dx}\sin(g) = \cos(g)\,g'$.
- Angles must be in radians.

The Big Picture

The trig derivatives

$\frac{d}{dx}\sin x = \cos x$ and $\frac{d}{dx}\cos x = -\sin x$ note the sign that appears with cosine.

Radians required

These formulas hold only when x is measured in **radians**, which is why calculus works in radian mode.

With the chain rule

For a composition, $\frac{d}{dx}\sin(g(x)) = \cos(g(x))\,g'(x)$.

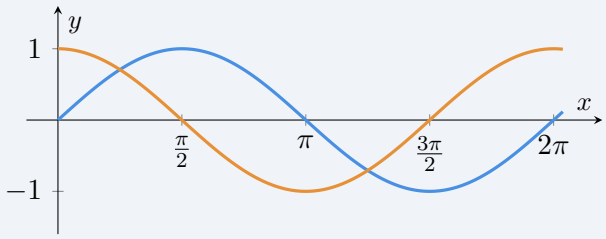
Worked Examples

Example 1 — Sine

Differentiate $f(x) = \sin x$.

Solution. **Step 1 (recall the basic trig derivative):** $\frac{d}{dx}\sin x = \cos x$.

Conclusion: $f'(x) = \cos x$ — the slope of sin is the height of cos:



Example 2 — Cosine

Differentiate $f(x) = \cos x$.

Solution. **Step 1 (recall the basic trig derivative):** $\frac{d}{dx}\cos x = -\sin x$.

Conclusion: $f'(x) = -\sin x$ (note the minus sign).

Example 3 — Chain rule

Differentiate $f(x) = \sin(3x)$.

Solution. **Step 1 (identify outer and inner):** outer is $\sin(\,\cdot\,)$, inner is $g(x) = 3x$ (so $g'(x) = 3$).
Step 2 (differentiate the outer, leaving the inner alone): $\cos(3x)$.
Step 3 (multiply by the inner derivative): $\cos(3x) \cdot 3$.
Conclusion: $f'(x) = 3\cos(3x)$.

Example 4 — Product rule

Differentiate $f(x) = x \sin x$.

Solution. **Step 1 (identify the two factors):** $u = x$ (so $u' = 1$) and $v = \sin x$ (so $v' = \cos x$).
Step 2 (apply the product rule $u'v + uv'$): $(1)\sin x + x\cos x$.
Conclusion: $f'(x) = \sin x + x\cos x$.

Example 5 — Power of sine

Differentiate $f(x) = \sin^2 x$.

Solution. **Step 1 (rewrite as a power of a function):** $f(x) = (\sin x)^2$, with inner $g(x) = \sin x$ (so $g'(x) = \cos x$).
Step 2 (differentiate the outer power, multiply by the inner derivative): $2(\sin x)^1 \cdot \cos x$.
Step 3 (recognize the double-angle identity, optional): $2\sin x \cos x = \sin 2x$.
Conclusion: $f'(x) = 2\sin x \cos x$ (equivalently $\sin 2x$).

Example 6 — Chain rule

Differentiate $f(x) = \cos(2x)$.

Solution. **Step 1 (identify outer and inner):** outer is $\cos(\,\cdot\,)$, inner is $g(x) = 2x$ (so $g'(x) = 2$).
Step 2 (differentiate the outer, using $\frac{d}{dx}\cos(u) = -\sin(u)$): $-\sin(2x)$.
Step 3 (multiply by the inner derivative): $-\sin(2x) \cdot 2$.
Conclusion: $f'(x) = -2\sin(2x)$.

Example 7 — Chain rule

Differentiate $f(x) = \sin(5x)$.

Solution. **Step 1 (identify outer and inner):** outer is $\sin(\,\cdot\,)$, inner is $g(x) = 5x$ (so $g'(x) = 5$).
Step 2 (differentiate the outer, multiply by the inner derivative): $\cos(5x) \cdot 5$.
Conclusion: $f'(x) = 5\cos(5x)$.

Example 8 — Product rule

Differentiate $f(x) = x^2 \cos x$.

Solution. **Step 1 (identify the two factors):** $u = x^2$ (so $u' = 2x$) and $v = \cos x$ (so $v' = -\sin x$).
Step 2 (apply the product rule): $2x\cos x + x^2(-\sin x)$.
Conclusion: $f'(x) = 2x\cos x - x^2 \sin x$.

Example 9 — Chain (square inside)

Differentiate $f(x) = \sin(x^2)$.

Solution. **Step 1 (identify outer and inner):** outer is $\sin(\,\cdot\,)$, inner is $g(x) = x^2$ (so $g'(x) = 2x$).

Step 2 (differentiate the outer, multiply by the inner derivative): $\cos(x^2) \cdot 2x$.

Conclusion: $f'(x) = 2x\cos(x^2)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = \cos(2x)$.

Question 2

Differentiate $f(x) = \sin(5x)$.

Question 3

Differentiate $f(x) = x^2 \cos x$.

Question 4

Differentiate $f(x) = \sin(x^2)$.

Question 5

Slope of $y = \sin x$ at $x = 0$?

Question 6

Differentiate $f(x) = 3\sin x$.

Question 7

Differentiate $f(x) = \cos(4x)$.

Question 8

Differentiate $f(x) = x\cos x$.

Question 9

Differentiate $f(x) = \cos^2 x$.

Question 10

Differentiate $f(x) = \sin(2x + 1)$.

Question 11

Slope of $y = \cos x$ at $x = \frac{\pi}{2}$?

Question 12

Differentiate $f(x) = \sin x + \cos x$.

Question 13 — Challenge

Differentiate $f(x) = \dfrac{\sin x}{x}$ (quotient rule).

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = \cos(2x)$.
2. Differentiate $f(x) = \sin(5x)$.
3. Differentiate $f(x) = x^2 \cos x$.
4. Differentiate $f(x) = \sin(x^2)$.
5. Slope of $y = \sin x$ at $x = 0$?
6. Differentiate $f(x) = 3\sin x$.
7. Differentiate $f(x) = \cos(4x)$.
8. Differentiate $f(x) = x\cos x$.
9. Differentiate $f(x) = \cos^2 x$.
10. Differentiate $f(x) = \sin(2x+1)$.
11. Slope of $y = \cos x$ at $x = \frac{\pi}{2}$?
12. Differentiate $f(x) = \sin x + \cos x$.
13. Differentiate $f(x) = \dfrac{\sin x}{x}$ (quotient rule).

✓ Answer Key

1. $-2\sin(2x)$
2. $5\cos(5x)$
3. $2x\cos x - x^2 \sin x$
4. $2x\cos(x^2)$
5. 1
6. $3\cos x$
7. $-4\sin(4x)$
8. $\cos x - x\sin x$
9. $-2\cos x \sin x = -\sin 2x$
10. $2\cos(2x+1)$
11. -1
12. $\cos x - \sin x$
13. $\dfrac{x\cos x - \sin x}{x^2}$